

1. (20 marks) For $(a, b, k) \in \mathbb{R}^3$, consider the following quadratic form

$$Q(\mathbf{x}) = ax_1^2 + kx_2^2 + bx_3^2 + 2ax_1x_2 + 2bx_2x_3.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a & a & 0 \\ a & k & b \\ 0 & b & b \end{pmatrix}$$

(2 marks) Its third order principal minor is $abk - a^2b - ab^2$. Its second order principal minors are $ak - a^2$, $bk - b^2$ and ab . Its first order principal minors are a , b , and k .

- (b) (2 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all parameter values for a , b , and k such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*?

Solutions: (2 marks) A quadratic form is indefinite if $Q(\mathbf{x}) > 0$ and $Q(\mathbf{y}) < 0$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$.

(4 marks) $Q(\mathbf{x})$ is positive semidefinite if all principal minors are positive which requires:

$$k \geq a + b \text{ and } \min\{a, b\} \geq 0.$$

The condition already accounts for scenarios in which some of the parameters are equal to 0.

(4 marks) $Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

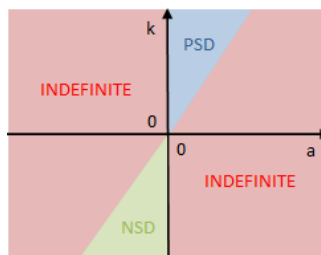
$$k \leq a + b \text{ and } \max\{a, b\} \leq 0.$$

The condition already accounts for scenarios in which some of the parameters are equal to 0.

(2 marks) $Q(\mathbf{x})$ is indefinite whenever it is not semidefinite.

- (c) (4 marks) For $b = a$, plot the pairs (a, k) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$.

Solutions: (4 marks) The plot below depicts the answer



2. (20 marks) Consider two *quasi-concave* functions $f : U \rightarrow \mathbb{R}$ and $g : U \rightarrow \mathbb{R}$ defined on some convex set U in $\mathbb{R}^n$.

- (a) (10 marks) Show that their sum,

$$S(\mathbf{x}) = f(\mathbf{x}) + g(\mathbf{x}) \text{ for all } \mathbf{x} \in U,$$

is not necessarily *quasi-concave* function. Then, establish that the sum S may even be *quasi-convex*.

Solutions: To show that S is not quasi-concave, consider the following example. Let $U = [0, 1]$ and let

$$\begin{aligned} f(x) &= \begin{cases} 1 & \text{if } x \leq 1/3 \\ 0 & \text{if } x > 1/3 \end{cases}, \\ g(x) &= \begin{cases} 1 & \text{if } x \geq 2/3 \\ 0 & \text{if } x < 2/3 \end{cases}. \end{aligned}$$

The upper-contour sets of these function are intervals and thus convex:

$$\begin{aligned} C_1^+(f) &= [0, 1/3] \quad \text{and} \quad C_0^+(f) = [0, 1], \\ C_1^+(g) &= [2/3, 1] \quad \text{and} \quad C_0^+(g) = [0, 1]. \end{aligned}$$

Thus, both functions are quasi-concave. Yet, their sum

$$S(x) = \begin{cases} 1 & \text{if } x \notin (1/3, 2/3) \\ 0 & \text{if } x \in (1/3, 2/3) \end{cases}$$

is not quasi-concave as one of its upper-contour sets is not convex:

$$C_1^+(S) = [0, 1/3] \cup [2/3, 1].$$

In the example considered, S is in fact quasi-convex. This follows because lower-contour sets are intervals and thus convex,

$$C_0^-(S) = (1/3, 2/3) \quad \text{and} \quad C_1^-(S) = [0, 1].$$

- (b) (5 marks) What are the implications of part (a) for an optimization problem with a *quasi-concave* the objective function and *quasi-convex* constraints? Discuss very briefly.

Solutions: The observation implies that the Lagrangian is not necessarily globally quasi-concave in $\mathbf{x}$ on the constraint set.

3. (40 marks) For $a > 0$, consider the problem:

$$3a \log(y+1) + x \quad \text{s.t.} \quad y + ax \leq 1, \quad y \geq 0, \quad x \geq a$$

- (a) (6 marks) Is the objective function concave? Are the constraints convex? For what values of $a > 0$ is Slater condition satisfied? So, for what values of $a > 0$, can you solve this optimization problem using the Kuhn-Tucker method?

Solutions: (3 marks) The objective function is concave, since its Hessian is negative semidefinite on the constraint set since for $y \geq 0$ and $x \geq a$

$$\begin{aligned} D^2 f(x, y) &= \begin{pmatrix} -3a/(y+1)^2 & 0 \\ 0 & 0 \end{pmatrix}, \\ |D^2 f(x, y)| &= 0 \geq 0, \quad -3a/(y+1)^2 \leq 0, \quad 0 \leq 0. \end{aligned}$$

The constraints are linear and thus convex. (3 marks) Thus, when the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case for all $a \in (0, 1)$, since the point $\tilde{x} = a + \varepsilon$ and $\tilde{y} = \varepsilon$ is always in the interior of the constraint set for $\varepsilon \in (0, (1 - a^2)/(1 + a))$, since

$$\tilde{y} + a\tilde{x} = \varepsilon + a(a + \varepsilon) < 1, \quad \tilde{x} > a, \quad \tilde{y} > 0,$$

which in turn requires $1 > a^2$ or equivalently $a < 1$.

- (b) (6 marks) Suppose the restrictions on a found in part (a) hold. Write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?

Solutions: (5 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= 3a/(y+1) - \lambda_1 + \lambda_2, \\ 0 &= 1 - a\lambda_1 + \lambda_3, \\ 0 &= \lambda_1(y+ax-1) = \lambda_2y = \lambda_3(x-a), \\ 0 &\geq y+ax-1, \quad 0 \geq -y, \quad 0 \geq a-x, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

(1 marks) Any solutions to these conditions is maximizer of the Lagrangian, because the Lagrangian is a sum of concave functions by the arguments in (a) and because Slater condition holds.

- (c) (20 marks) Suppose the restrictions on a found in part (a) hold. Solve the constrained optimization problem in terms of a .

Solutions: Consider the various supports:

- (1 marks) All three multipliers cannot be equal to 0, since FOC would be violated given the objective is increasing in both variables.
- (1 marks) All three multipliers can be strictly positive only if $a = 1$. If so, the three complementary slackness conditions would be met by setting $x(a) = a$ and $y(a) = 0$. Otherwise, one of complementary slackness conditions would necessarily be violated for any $a \neq 1$.
- (6 marks) If $\lambda_2(a) = 0$ and $\lambda_1(a), \lambda_3(a) > 0$, then by complementary slackness $x(a) = a$ and $y(a) = 1 - a^2 \geq 0$ for $a \leq 1$. Further, FOC would then imply that $\lambda_1(a) = 3a/(2 - a^2) > 0$ and

$$\lambda_3(a) = \frac{3a^2}{2 - a^2} - 1 \geq 0 \quad \Leftrightarrow \quad 2a^2 \geq 1.$$

So as multipliers are non-negative, this is a solution for $a \in [1/\sqrt{2}, 1)$.

- (6 marks) If $\lambda_3(a) = 0$ and $\lambda_1(a), \lambda_2(a) > 0$, then by complementary slackness $y(a) = 0$ and $x(a) = 1/a > 0$ for $a > 0$. Further, FOC would then imply that $\lambda_1(a) = 1/a > 0$ and

$$\lambda_2(a) = \frac{1}{a} - 3a \geq 0 \quad \Leftrightarrow \quad 3a^2 \leq 1.$$

So as multipliers are non-negative, this is a solution for $a \in (0, 1/\sqrt{3}]$.

- (6 marks) Finally, if $\lambda_3(a) = \lambda_2(a) = 0$ and $\lambda_1(a) > 0$, then FOC and the constraints would imply that

$$\begin{aligned}\lambda_1(a) &= \frac{1}{a} \geq 0, \\ y(a) &= 3a^2 - 1 \geq 0 \Leftrightarrow 3a^2 \geq 1, \\ x(a) &= (2 - 3a^2)/a \geq a \Leftrightarrow 1 \geq 2a^2.\end{aligned}$$

So as constraints need to be satisfied, this is a solution for $a \in [1/\sqrt{3}, 1/\sqrt{2}]$

- (d) (8 marks) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to find the slope of value function with respect to the parameter a . Is there a value for a that maximizes the value function $v(a)$ of this problem?

Solutions: The Envelope Theorem states that if $(x(a), y(a), \lambda(a))$ are differentiable functions and the constraint qualification holds, then

$$\frac{dv(a)}{da} = \frac{\partial L(x(a), y(a), \lambda(a))}{\partial a} = 3 \log(y(a) + 1) - \lambda_1(a)x(a) - \lambda_3(a).$$

(2 marks) If $a \in (0, 1/\sqrt{3}]$, then

$$\frac{dv(a)}{da} = -\frac{1}{a^2}.$$

Intuitively, the slope of the value function is decreasing as the constraint set shrinks and $y(a)$ is set to 0.

(2 marks) If $a \in [1/\sqrt{3}, 1/\sqrt{2}]$, then

$$\frac{dv(a)}{da} = 3 \log(3a^2) - \frac{2}{a^2} + 3.$$

The slope still decreases until $a = 0.69636$ and then increases. The reduction in the constraint set eventually gets compensated by the increase in the objective.

(2 marks) Finally, if $a \in [1/\sqrt{2}, 1)$, then

$$\frac{dv(a)}{da} = \frac{\partial L(x(a), y(a), \lambda(a))}{\partial a} = 3 \log(2 - a^2) - \frac{6a^2}{2 - a^2} + 1.$$

The slope still increases until $a = 0.72733$ and then decreases.

(2 marks) There is no value of a maximizing the value function, as

$$\lim_{a \rightarrow 0} v(a) = \infty$$

given that $\lim_{a \rightarrow 0} x(a) = \lim_{a \rightarrow 0} 1/a = \infty$.

4. (20 Marks) Consider the correspondence $\mathbf{h} : [0, 1]^2 \rightrightarrows [0, 1]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_2 \in (3/4, 1] \\ 2x_2 - 1/2 & \text{if } x_2 \in [1/4, 3/4] \\ 0 & \text{if } x_2 \in [0, 1/4) \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [0, a] & \text{if } x_1 = 1 \\ a & \text{if } x_1 \in (0, 1) \\ [0, a] & \text{if } x_1 = 0 \end{cases},$$

for a scalar $a \in [0, 1]$. Briefly, argue why *Kakutani fixed point theorem* applies. Then, for all values of $a \in [0, 1]$ find all the *fixed points* of this correspondence.

Solutions: (2 marks) Both correspondences have closed graph and are therefore UHC. Moreover, both are obviously convex-valued and non-empty. Since the image of $\mathbf{h}$ belongs to $[0, 1]^2$ which is also the domain, the Fixed Point Theorem applies. Thus, there will be at least one fixed point on $[0, 1]^2$.

(6 marks) First suppose that $a \leq 1/4$. If so, the set of fixed points of $\mathbf{h}$ consists of points $(0, x_2)$ for all $x_2 \leq a$. The points $(0, x_2)$ for any $x_2 \in [0, a]$ are fixed points since

$$(0, x_2) \in \{0\} \times [0, a] = \mathbf{h}(0, x_2).$$

Moreover, no other point can be a fixed point if $a < 1/4$, because $h_2(\mathbf{x}) \leq a$ and $h_1(\mathbf{x}) = 0$ whenever $x_2 \leq a \leq 1/4$.

(6 marks) Next suppose that $a \in (1/4, 3/4)$. If so, the set of fixed points of $\mathbf{h}$ consists of $(0, x_2)$ for all $x_2 \leq 1/4$ and $(2a - 1/2, a)$. The points $(0, x_2)$ for any $x_2 \in [0, 1/4]$ are fixed points since

$$(0, x_2) \in \{0\} \times [0, a] = \mathbf{h}(0, x_2).$$

The point $(2a - 1/2, a)$ is a fixed point since for $a \in (1/4, 3/4)$

$$(2a - 1/2, a) \in \{2a - 1/2\} \times \{a\} = \mathbf{h}(2a - 1/2, a).$$

Moreover, no other point can be a fixed point because: (i) $h_1(\mathbf{x}) = 0$ for $x_2 \leq 1/4$ and so $h_2(\mathbf{x}) \leq 1/4$; and (ii) $h_1(\mathbf{x}) = 2x_2 - 1/2$ for $x_2 \in (1/4, a)$ and $h_2(\mathbf{x}) \leq a$.

(6 marks) Finally, suppose that $a \in [3/4, 1]$. If so, the set of fixed points of $\mathbf{h}$ consists of $(0, x_2)$ for all $x_2 \leq 1/4$ and $(1, x_2)$ for all $x_2 \in [3/4, a]$. The points $(0, x_2)$ for any $x_2 \in [0, 1/4]$ are fixed points since

$$(0, x_2) \in \{0\} \times [0, a] = \mathbf{h}(0, x_2).$$

The points $(1, x_2)$ for any $x_2 \in [3/4, a]$ are fixed points since

$$(1, x_2) \in \{1\} \times [0, a] = \mathbf{h}(1, x_2).$$

Moreover, no other point can be a fixed point because: (i) $h_2(\mathbf{x}) \leq a$; and (ii) $x_1 \in (0, 1)$ would imply $h_2(\mathbf{x}) = a$ which would in turn imply $h_1(\mathbf{x}) = 1$ for $a \geq 3/4$.